David Ho

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Relevant Skills

Languages: Python, TypeScript, JavaScript, C++, Java, HTML/CSS, C#
Frameworks: React, Next.js, Express, Flask, Tailwind CSS, PyTorch, TensorFlow, SQLAlchemy, Pandas, Pytest
Data & Storage: PostgreSQL, MySQL, SQLite, Redis, DynamoDB, Pinecone, AWS S3, Alembic
Other: OpenAI API, LangChain, Node.js, Git, Docker, Google Colab, LaTeX, AWS Lambda, CloudFront, Vercel

Projects

Credit Card Rewards Optimizer
- Built a full-stack web app to recommend optimal credit cards for purchases using category-based and semantic search.
- Designed an embedding-based engine (Pinecone + OpenAI with GPT fallback) with a React/Next.js + Tailwind frontend and an Express/Node.js + PostgreSQL backend securely deployed on Vercel.

RAG-Based Recipe Generator
- Built a full-stack serverless recipe generation app using OpenAI, Pinecone, and LangChain for RAG-based retrieval.
- Deployed frontend (React/TypeScript) via AWS S3 + CloudFront; backend (Lambda + DynamoDB) on AWS.

Sudoku Web Application with AI Hints
- Built a full-stack web app with React, TypeScript, and Tailwind CSS, with OpenAI GPT API integration for hints.
- Implemented a Flask backend with PostgreSQL for user authentication and data persistence, deployed on Vercel.

Experience

Holor Technology, Inc. **Jun 2024 – Present**
R&D Software Engineer *(Remote) Yokohama, Japan*
- Built full-stack web platform replacing SSH workflows for on-premise electromagnetic simulation servers, enabling authenticated job submission and persistent tracking of 20+ minute runs (React, TypeScript, FastAPI, PostgreSQL)
- Reduced end-to-end simulation testing time by 15% through automated job management and streamlined result retrieval, eliminating manual server access for engineering team
- Collaborated with a team of three engineers in a fast-paced startup to design a novel electromagnetic simulation method using the velocity gauge, improving accuracy and efficiency.
- First author of *Numerical Methods for Electromagnetic Simulation using Velocity Gauge*, submitted to the 62nd DAC.

Holor Technology, Inc. **Jun – Aug 2023**
Application Engineer *(Remote) Yokohama, Japan*
- Created a user manual from scratch in LaTeX for a 3D parasitic extraction and simulation software, enhancing user understanding and operational efficiency.
- Developed and documented comprehensive procedures for ECO simulation and layout traceability, leading to a 10% improvement in design validation accuracy, while enhancing user efficiency by simplifying complex workflows.

Holor Technology, Inc. **Jun – Aug 2022**
Software Engineer Intern *(Remote) Yokohama, Japan*
- Conducted data analysis with Pandas, and created Matplotlib visualizations to analyze trends and uncover bugs and unintended results, processed and cleaned datasets, supporting research on a novel numerical method and improving accuracy by 15%.

Google **Sep – Dec 2021**
CS Research Mentorship Program Intern *Remote*
- Designed and implemented machine learning models, including linear regression using PyTorch Autograd, to analyze and optimize real-world datasets, applying gradient-based optimization techniques.
- Conducted in-depth research on topics in Physics-Based Deep Learning, synthesizing key insights from papers and presenting findings to technical teams.

Stanford University Engineering Department **Jun – Jul 2016**
Machine Learning Intern *Palo Alto, CA*
- Implemented edge detection and CNN algorithms using Deeplearning4j for image recognition, improving accuracy and reducing processing time by 20%.
- Developed a fully-connected neural network in Java to classify handwritten digits from a dataset of 60,000 images, with Arduino Wi-Fi boards for hardware connectivity, achieving an accuracy of 98%.

Education

- **University of Illinois at Urbana-Champaign:** *Bachelor of Science: Engineering Physics, Minor: Computer Science*